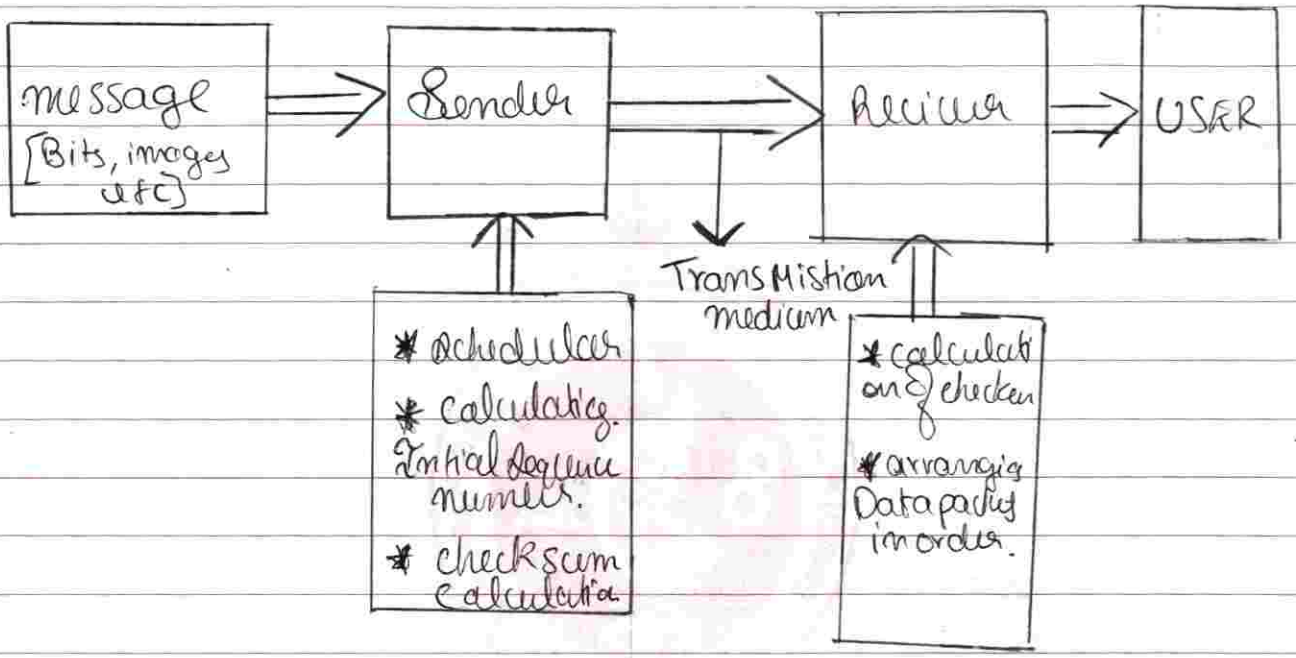
Q. Nos.

* Protocol: This is the set of the rules that must have to be followed inorder to transmit the data without following the protocal the transfered data is just an ordinary electronic signals which is of no use.



The three criteria's necessary for an effective & efficient network.,

* Error detation:- due to the noise in the trans-misstion medium the Data packets get corrupted and cause the error or Damages the Data to avoid this acceptance & the corrupted data, establish the virtual connection circuit where the Data packets dosnot miss and calculate the checksum on each Data packets and mention ip adresses & the source & destination.

07

Q. Nos.

* By dividing the network into layers:- By the division of the network into layers, where each layer works on a specific task & improves the efficiency.

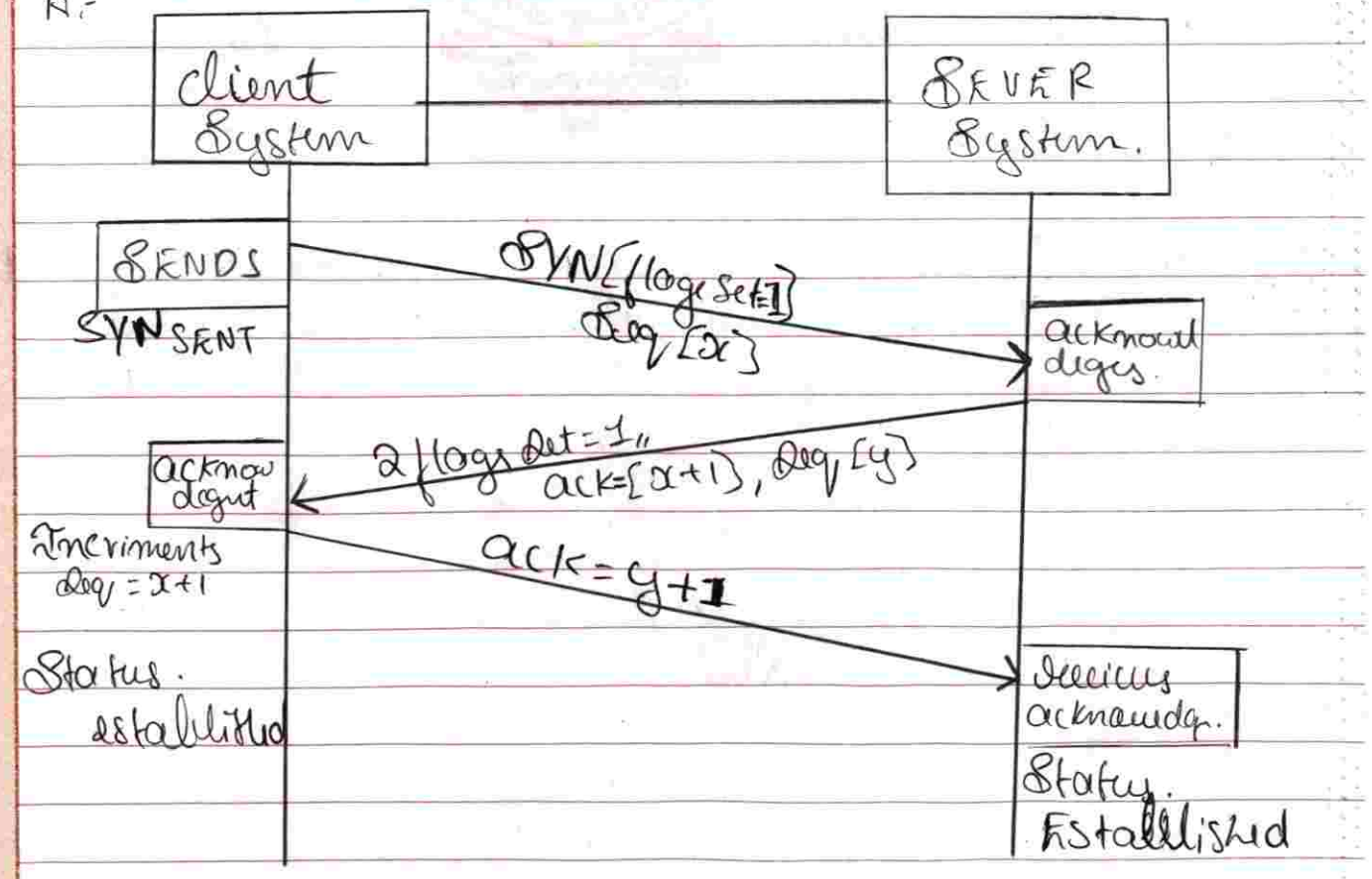
* Bidirectional:- In the transmittion medium it should consist of Bidirectional property where each layer should be ready to perform the one complimentary tasks.

Ex:- the nodes most be ready for either recieving and sending the Data at Both the ends.

Module-4

Q8 a)

A:-



Q. Nos.

TCP Connection Establishment:-

The Connection is being established through a process called as three-way handshake. TCP.

* Where here the Sender from the client side initiate the transmission By generating a Control Segment with SYN[Synchronization flag] Set to 1 & gives randomly a Initial Sequence number to its Data Stream. Set it be Seq = x, Status is said to be SYN SENT.

* On receiving side the SERVER System ~~for~~ receives the this SYN flag and Seq & INS[Initial Sequence number] and generate a buffer memory for it.

* Then On receiving side. the two flages are set to 1 ACK = x+1 & SYN = Y, and sent to the client System. the Status is acknowledgement sent.

* On client side it receives the ~~ack~~ ACK & SYN and increments its Stream to Seq = x+1 and Sends Back a acknowledgement ack = y+1 to the SERVER System.

* After receiving this ACK, the SERVER System Establish the TCP connection and now on Both a Side the connection is being established.

* The Transmission of the Data Can be done.

* After the transmission of the Data, the connection is to be terminated.

* Where again the client sends a SYN to END the the connection. Stays wait.
SEVER

* And the System goes through this SYN and gives an acknowledgement. Back to the client and the System Stays of the System is wait.

* Then again an acknowledgent is being sent By the client to SEVER System. at this point the connection is almost closed. no Data transfer is happening.

* AND the client looks once with the transmission medium whether any data is left and empty it.

* the connection can't be terminated immediately after the acknowledgment it takes time of.
2 MSS 2 MSS.

* This is called as 4 way handShake which is involved in the termination of the connection.

ii) Analyzing the duration if one of the handshake. packets is lost or delayed.

* if any of the packets is lost while transmission sequential transmission is happening.

Q. Nos.

* at a receiver side while transmission the receiver sends an acknowledgment for each Data packets it receives.

* if a packet is being missed also the receiver sends a Duplicate ack to order to re transmit the specific Data packets.

* the lost Data packet is easily recovered due to the initial sequence numbers of the Data packets which helps to arrange them in a perfect sequential order.

Q8

Answer:

A:-

Flow control & Congestion Control mechanism. Used in TCP.

In TCP[Transport layer protol] the problem araise when the Data packets gets collision in between them and lose of packets or the Corruption of the Data packets happens due to this In the transmission medium. So in order to control this collision and fullfill the Draw Backs the Flow control and the congestion control mechanism are being introduced.